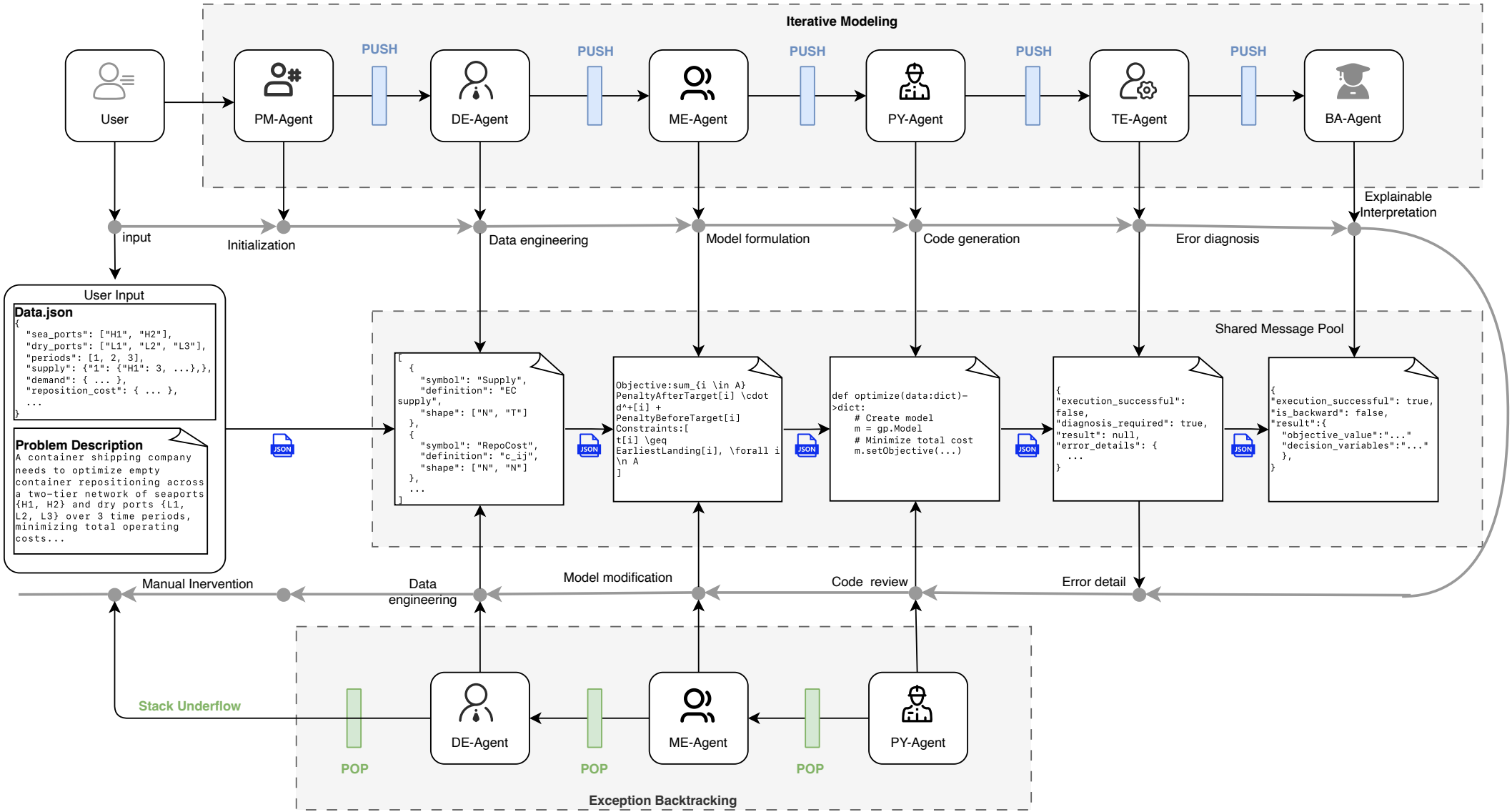
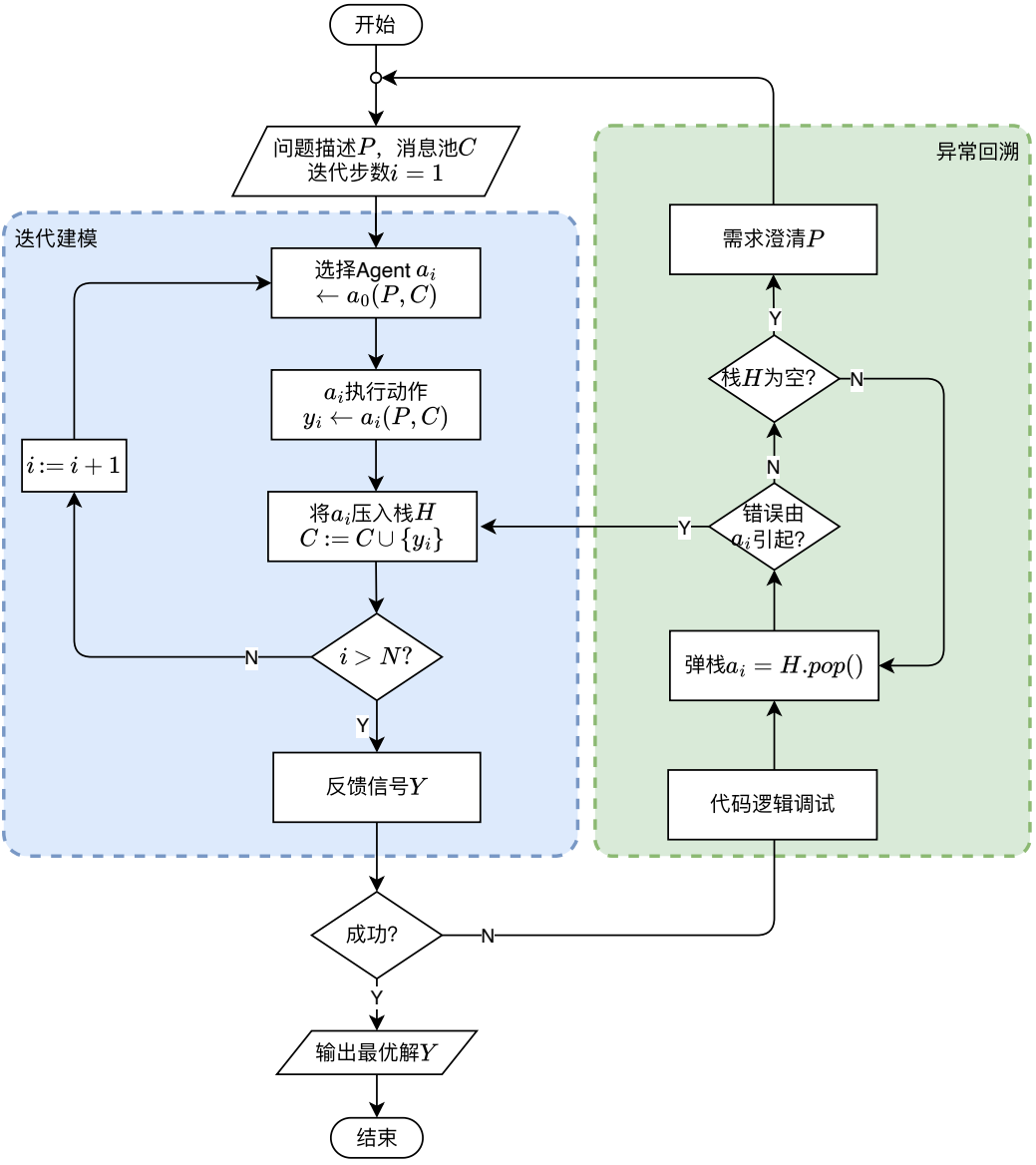
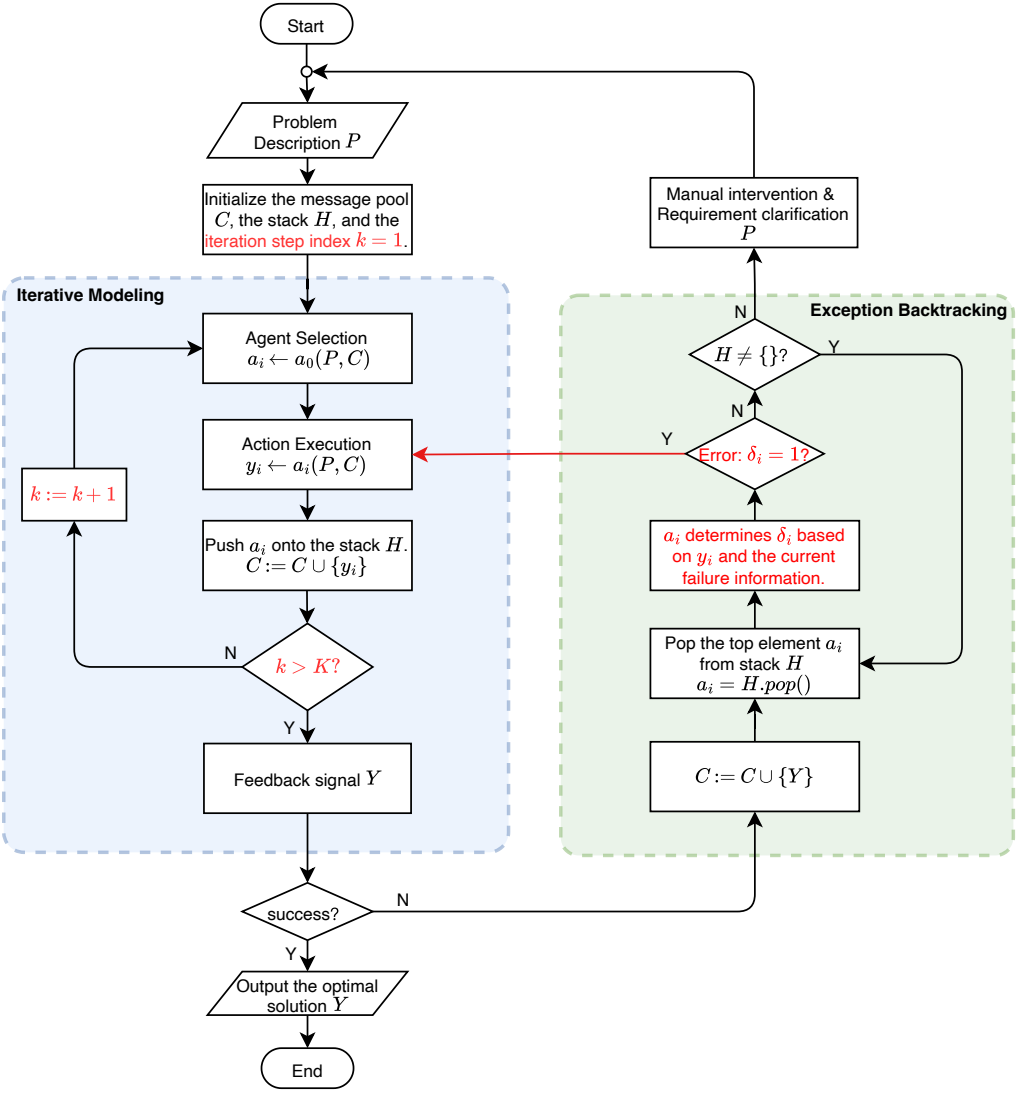

Iterative Modeling
User
PM-Agent
DE-Agent
ME-Agent
PY-Agent
TE-Agent
BA-Agent
PUSH
PUSH
PUSH
PUSH
PUSH
input
Initialization
Data engineering
Model formulation
Code generation
Eror diagnosis
Explainable Interpretation
User Input
Data.json
{
"sea_ports": ["H1", "H2"],
"dry_ports": ["L1", "L2", "L3"],
"periods": [1, 2, 3],
"supply": {"1": {"H1": 3, ...},},
"demand": { ... },
"reposition_cost": { ... },
...
}
Problem Description
A container shipping company needs to optimize empty container repositioning across a two-tier network of seaports {H1, H2} and dry ports {L1, L2, L3} over 3 time periods, minimizing total operating costs...
Shared Message Pool
[
{
"symbol": "Supply",
"definition": "EC supply",
"shape": ["N", "T"]
},
{
"symbol": "RepoCost",
"definition": "c_ij",
"shape": ["N", "N"]
},
...
]
Objective:sum_{i \in A} PenaltyAfterTarget[i] \cdot d^+[i] + PenaltyBeforeTarget[i]
Constraints:[
t[i] \geq EarliestLanding[i], \forall i \n A
]
def optimize(data:dict)->dict:
# Create model
m = gp.Model
# Minimize total cost
m.setObjective(...)
{
"execution_successful": false,
"diagnosis_required": true,
"result": null,
"error_details": {
...
}
{
"execution_successful": true,
"is_backward": false,
"result":{
"objective_value":"..."
"decision_variables":"..."
},
}
JSON
Manual Inervention
Data engineering
Model modification
Code review
Error detail
Stack Underflow
DE-Agent
ME-Agent
PY-Agent
POP
POP
POP
Exception Backtracking


开始
问题描述P，消息池C
迭代步数$i = 1$
迭代建模
选择Agent a_i
$\leftarrow a_0(P, C)$
a_i执行动作
$y_i \leftarrow a_i(P, C)$
将a_i压入栈H
$C := C \cup \{y_i\}$
$i > N$?
N
$i := i + 1$
Y
反馈信号Y
成功?
N
Y
输出最优解Y
结束
异常回溯
代码逻辑调试
弹栈$a_i = H.pop()$
错误由
a_i引起?
Y
N
栈H为空?
N
Y
需求澄清P


Start
Problem Description P
Initialize the message pool C, the stack H, and the iteration step index $k = 1$.
Iterative Modeling
Agent Selection $a_i \leftarrow a_0(P, C)$
Action Execution $y_i \leftarrow a_i(P, C)$
Push a_i onto the stack H. $C := C \cup \{y_i\}$
$k > K$?
N
Y
$k := k + 1$
Feedback signal Y
success?
N
Y
Output the optimal solution Y
End
Manual intervention & Requirement clarification P
Exception Backtracking
$H \neq \{\}$?
N
Y
Error: $\delta_i = 1$?
N
Y
a_i determines δ_i based on y_i and the current failure information.
Pop the top element a_i from stack H $a_i = H.pop()$
$C := C \cup \{Y\}$